DSC5001: Statistical Machine Learning I

Week 8: Classification I: Discriminant Analysis and Logistic Regression

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Semester A, 2026/27

Assignment 2

Assignment 2 is now available on Canvas.

- **Deadline:** November 5, 2026 (unchanged).
- You have a longer period to complete the assignment.
- This is a **written assignment**.
- No programming or coding submission is required.

Learning Objectives

By the end of this lecture, you should be able to:

1. turn class probabilities into decisions and evaluate the errors;
2. explain and apply LDA and QDA;
3. compute logistic probabilities and explain how logistic regression is trained;
4. interpret thresholds, ROC/PR measures, and probability calibration.

Roadmap

Part 1 Classification, probabilities, and decision costs

Part 2 Gaussian discriminant analysis: LDA and QDA

Part 3 Logistic regression: probability, loss, and training

Part 4 Thresholds, evaluation, and probability quality

Part 5 Choosing a classifier

Main idea

A good classifier needs useful probabilities *and* a decision rule suited to the task.

Classification decisions

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Classification Predicts Categories

Regression predicts a numerical outcome. Classification predicts a category.
For medical screening, define

$$Y = 0 \text{ (negative)}, \qquad Y = 1 \text{ (positive)}.$$

From patient features $\boldsymbol{x}$, such as test measurements, a classifier can estimate

$$\boxed{p(\boldsymbol{x}) = \mathbb{P}(Y = 1 \mid X = \boldsymbol{x}).}$$

Two steps: estimate the probability, then choose a class using a decision rule.

Training Data and Prediction Notation

We observe n labeled examples

$$\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}.$$

- $\boldsymbol{x}_i$: feature vector of observation i.
- y_i: its true label, e.g., 0 or 1.
- p: number of input features in each $\boldsymbol{x}_i$.
- K: number of possible classes, labeled $1, \ldots, K$ for multiclass problems.

For a new input $\boldsymbol{x}$, the predicted label is $\widehat{y}$; its estimated class probabilities are $\widehat{p}_k(\boldsymbol{x})$.

Probabilities Carry More Information

Two patients receive the same positive decision at threshold $t = 0.5$.

	Patient A	**Patient B**
Estimated disease probability	0.52	0.95
Predicted class	1	1

The class labels are identical, but the estimated risks differ greatly.

Main idea

Probabilities are useful when decisions must reflect different error costs.

Bayes Decision Under Equal Costs

Suppose the true class is one of $1, \ldots, K$, and every mistake costs the same.
If we predict class k, the probability of being wrong is

$$\boxed{R(k \mid \boldsymbol{x}) = 1 - \mathbb{P}(Y = k \mid X = \boldsymbol{x}).}$$

We therefore choose the class with the largest posterior probability:

$$\boxed{g^*(\boldsymbol{x}) = \arg\max_k \mathbb{P}(Y = k \mid X = \boldsymbol{x}).}$$

Example: if the class probabilities are $(0.15, 0.55, 0.30)$, choose class 2; its conditional error is $1 - 0.55 = 0.45$.

Different Errors Have Different Costs

Setting: a screening test can make two types of mistakes. Missing a serious illness may be more costly than a false alarm.
Let $p = \mathbb{P}(Y = 1 \mid \boldsymbol{x})$ be the patient's disease probability.

- False alarm (predict 1, actual 0): cost 1.
- Missed disease (predict 0, actual 1): cost 10.
- Correct decision: cost 0.

The expected costs are

Predict positive:	$1(1-p)$,
Predict negative:	$10p$.

Choose the decision with the smaller expected cost.

Decision Costs: Numerical Example

Using false-positive cost 1 and false-negative cost 10, predict positive when

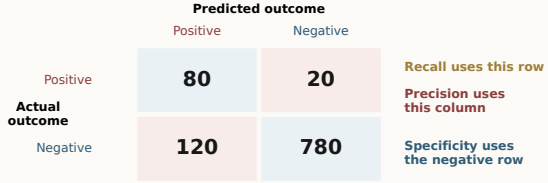
$$1(1-p) < 10p \quad \Longleftrightarrow \quad \boxed{p > 1/11 \approx 0.091.}$$

	Patient A	**Patient B**
Estimated disease probability	0.18	0.05
Decision under these costs	positive	negative

Main idea

The threshold changes because the costs change. The probability model itself does not change.

Classification Counts



Actual label
What happened

Model decision
What we predicted

Outcome counts
TP, FP, TN, FN

Takeaway. Each evaluation measure uses a particular group of these cases.

Four Classification Outcomes

Use class 1 as the positive class.

	Predicted 1	**Predicted 0**
Actual 1	TP	FN
Actual 0	FP	TN

- **TP**: detected an actual positive; **FN**: missed an actual positive.
- **FP**: false alarm; **TN**: correctly rejected an actual negative.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Sensitivity, Specificity, and Precision

The same confusion matrix supports several measures. Each has a different denominator.

Measure	Formula	Meaning
Recall (sensitivity)	$\frac{TP}{TP+FN}$	Share of actual positives detected
Specificity	$\frac{TN}{TN+FP}$	Share of actual negatives rejected
Precision	$\frac{TP}{TP+FP}$	Share of positive predictions correct

Common mistake

Recall and precision answer different questions. Do not exchange their denominators.

Confusion-Matrix Example

Suppose 100 screening decisions produce

$$TP = 16, \quad FN = 4, \quad FP = 6, \quad TN = 74.$$

Measure	Calculation	Result
Accuracy	$90/100$	0.900
Recall	$16/20$	0.800
Specificity	$74/80$	0.925
Precision	$16/22$	0.727

The model is correct in 90% of cases, but misses 4 of the 20 actual positives.

Discriminant analysis

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Motivation for Discriminant Analysis

Setting: a clinic has measurements from healthy and diseased patients. We want to classify a new patient's measurements.
Instead of estimating the disease probability separately at every possible feature value, we can learn:

1. how common each class is;
2. the typical measurements within each class;
3. which class better explains a new measurement.

Discriminant analysis combines those descriptions using Bayes' rule to obtain class probabilities.

Class Priors and Posterior Probabilities

For a class k, distinguish three quantities:

- $\pi_k = \mathbb{P}(Y = k)$: **prior** class frequency;
- $f_k(\boldsymbol{x}) = p(\boldsymbol{x} \mid Y = k)$: **density** of the features within class k;
- $p_k(\boldsymbol{x}) = \mathbb{P}(Y = k \mid X = \boldsymbol{x})$: **posterior** class probability.

Bayes' rule combines them:

$$\boxed{p_k(\boldsymbol{x}) = \frac{\pi_k f_k(\boldsymbol{x})}{\sum_{j=1}^{K} \pi_j f_j(\boldsymbol{x})}.}$$

Here K is the number of classes; the denominator normalizes their probabilities.

Bayes' Rule: Email Classification

Suppose 10% of incoming emails are spam, so

$$\mathbb{P}(\text{spam}) = 0.10.$$

An email contains the word *urgent*. Suppose

$$\mathbb{P}(\text{urgent} \mid \text{spam}) = 0.60, \quad \mathbb{P}(\text{urgent} \mid \text{not spam}) = 0.10.$$

Then

$$\boxed{\mathbb{P}(\text{spam} \mid \text{urgent}) = \frac{0.60(0.10)}{0.60(0.10) + 0.10(0.90)} = 0.40.}$$

The word is informative, but the uncommon class still has posterior probability below 0.5.

Motivation for Gaussian Class Models

Bayes' rule needs to describe which feature values are likely *within each class*.
With limited observations, it is hard to estimate an unrestricted distribution. A Gaussian model uses only:

- a mean $\boldsymbol{\mu}_k$: the class center;
- a covariance $\boldsymbol{\Sigma}_k$: the class spread and feature relationships.

Main idea

LDA shares covariance across classes. QDA allows different covariance for each class.

Gaussian Distributions Within Classes

For one measurement X, a class-specific Gaussian model is

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \sigma_k^2).$$

- μ_k: mean measurement in class k;
- σ_k: standard deviation within that class;
- σ_k^2: variance.

For example, if $\mu_k = 50$ and $\sigma_k = 10$, then $x = 60$ lies one standard deviation above the class mean.

A larger distance from the mean generally gives a smaller Gaussian density, with the spread also affecting the comparison.

LDA Uses Shared Variance

Linear Discriminant Analysis (LDA) describes classes with different centers but a common spread.
For one predictor, it assumes

$$\boxed{X \mid Y = k \sim \mathcal{N}(\mu_k, \sigma^2),}$$

where μ_k depends on the class and σ^2 is shared.

Main idea

Sharing the variance uses fewer parameters. It can be more stable when each class has relatively few examples.

LDA Example: Patient Classification

Suppose a biomarker follows these illustrative distributions:

	Healthy ($Y = 0$)	**Disease ($Y = 1$)**
Class mean μ_k	40	60
Std. deviation σ	10	10
Class prior π_k	0.5	0.5

A new patient has value $x = 55$.
Goal: compare how likely this value is under the two Gaussian class models.

LDA Example: Compare Class Scores

For $x = 55$, the class means are 40 and 60, and both variances are 100.
Equal priors and equal variances contribute the same terms to both scores. Compare only the remaining terms:

$$\delta_0 = -\frac{(55-40)^2}{2(100)} = \boxed{-1.125},$$

$$\delta_1 = -\frac{(55-60)^2}{2(100)} = \boxed{-0.125}.$$

Since $\delta_1 > \delta_0$, choose $\boxed{\text{class } 1}$.
These are **reduced log scores**, not probabilities.

LDA Example: Posterior Probability

Same patient: the two scores are -1.125 and -0.125.
Exponentiate and normalize the scores to recover the posterior:

$$\boxed{p_1(55) = \frac{e^{-0.125}}{e^{-1.125} + e^{-0.125}} = \frac{1}{1 + e^{-1}} \approx 0.731.}$$

The LDA model assigns about 73.1% probability to class 1.

Main idea

A discriminant score ranks classes; the normalized posterior expresses class probabilities.

Class Priors Affect the Posterior

Now suppose disease is less common, with priors $\pi_1 = 0.20$ and $\pi_0 = 0.80$.
At biomarker value $x = 53$, the class-conditional density ratio is

$$\frac{f_1(53)}{f_0(53)} = e^{0.6} \approx 1.82.$$

Using Bayes' rule,

$$\boxed{p_1(53) = \frac{0.20e^{0.6}}{0.80 + 0.20e^{0.6}} \approx 0.313.}$$

The measurement favors the disease distribution, but the lower disease frequency keeps the posterior below 0.5.

One-Dimensional LDA Score

With a common variance σ^2, Gaussian terms shared by all classes can be removed.
One useful LDA class score is

$$\boxed{\delta_k(x) = \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.}$$

- x: new measurement; μ_k: class mean;
- π_k: prior class probability; σ^2: shared variance.

Choose the class with the largest score. The score itself is not a probability.

Linear Decision Boundaries in LDA

A decision boundary is where two classes receive equal scores.
For the equal-prior LDA example with means 40 and 60, equal scores occur at

$$\boxed{x = 50.}$$

Scores on one side favor class 0; scores on the other favor class 1.
With multiple features, the shared covariance makes the score difference **linear in the input**.
Its boundary is a line or hyperplane.

Multivariate LDA Score

For p features, let $\boldsymbol{x} \in \mathbb{R}^p$ be a new input, $\boldsymbol{\mu}_k \in \mathbb{R}^p$ the mean of class k, and $\boldsymbol{\Sigma}$ the covariance shared by all classes.
The LDA score is

$$\delta_k(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - \tfrac{1}{2} \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k + \log \pi_k.$$

$\boldsymbol{\Sigma}^{-1}$ adjusts for feature scales and correlations; π_k is the class prior. Choose the class with the largest score.

Estimating LDA Parameters

LDA estimates three quantities from the labeled training set:

1. **Class frequencies:** if 30 of 50 observations are class 0, then $\widehat{\pi}_0 = 30/50 = 0.60$.
2. **Class means:** average the feature vectors separately within each class.
3. **Shared covariance:** combine deviations around *each class's own mean*.

Main idea

Pooling within-class variability gives one covariance estimate. Mixing the class means into this spread estimate would distort the model.

Motivation for QDA

Limitation of LDA: two groups may have quite different spreads or feature correlations. If LDA forces them to share one covariance, it can miss important differences between the groups.
Quadratic Discriminant Analysis (QDA) instead assumes

$$\boxed{\boldsymbol{X} \mid Y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),}$$

where class k has its own covariance $\boldsymbol{\Sigma}_k$.
More flexibility requires more observations in each class to estimate covariance reliably.

QDA Discriminant Score

QDA uses a separate mean $\boldsymbol{\mu}_k$, covariance $\boldsymbol{\Sigma}_k$, and prior π_k for each class.
The class score is

$$\boxed{\begin{aligned}\delta_k(\boldsymbol{x}) = & -\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| \\ & -\tfrac{1}{2}d_k^2(\boldsymbol{x}) + \log\pi_k,\end{aligned}}$$

where

$$d_k^2(\boldsymbol{x}) = (\boldsymbol{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k).$$

The distance term measures how unusual the input is within class k; the determinant term accounts for its spread.

QDA Example: Unequal Variances

Consider two equally common classes with the *same* mean but different variances:

$$X \mid Y = 0 \sim \mathcal{N}(0, 1), \qquad X \mid Y = 1 \sim \mathcal{N}(0, 4).$$

After dropping constants common to both classes,

$$\delta_0(x) = -\frac{x^2}{2}, \qquad \delta_1(x) = -\frac{1}{2}\log 4 - \frac{x^2}{8}.$$

At $x = 0$: $\delta_0 = 0 > \delta_1 \approx -0.693$, so choose $\boxed{0}$.
At $x = 3$: $\delta_0 = -4.5 < \delta_1 \approx -1.818$, so choose $\boxed{1}$.

Unequal Variances Change Predictions

In the QDA example, both class means are zero, but class 1 is more spread out.

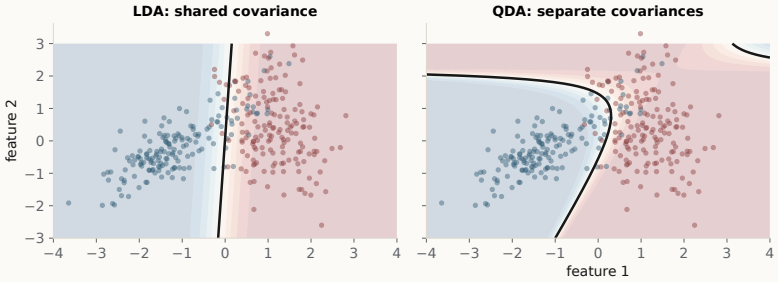
Input value	Class with larger density
Near zero	0 (narrow distribution)
Far from zero	1 (wide distribution)

With equal priors, the score equality occurs at approximately $x = \pm 1.36$.

Main idea

A QDA decision can change at two cutoffs because class variances differ.

LDA and QDA Boundaries



Dots

labeled observations

LDA

straight boundary

QDA

potentially curved boundary

Takeaway. The difference in boundary shape comes from the covariance assumptions.

LDA and QDA Compared

	LDA	QDA
Covariance	one shared matrix	separate for each class
Decision boundary	linear	can be curved
Number of parameters	fewer	more
Useful when	class data are limited	class spreads differ

Main idea

A more flexible boundary is not automatically better. Compare validation performance.

Logistic regression

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Motivation for Logistic Regression

Motivation: LDA and QDA first model how measurements are distributed *within* each class. Sometimes we only need the class-one probability and do not want to assume Gaussian feature distributions.
Logistic regression models directly

$$\boxed{p(\boldsymbol{x}) = \mathbb{P}(Y = 1 \mid X = \boldsymbol{x}).}$$

It is a useful, interpretable baseline for binary labels $Y \in \{0, 1\}$.

Linear Scores and Valid Probabilities

Suppose we predict the chance of passing an exam from hours of study.
An unrestricted straight line can give outputs such as -0.2 or 1.3, which cannot be probabilities.
Solution: compute a linear score and then convert that score to the interval $(0, 1)$ using a **sigmoid**.

$$\boxed{\text{features} \longrightarrow \text{linear score} \longrightarrow \text{probability.}}$$

Sigmoid Transforms a Score

The sigmoid, also called the logistic function, is

$$\boxed{\sigma(z) = \frac{1}{1 + e^{-z}}.}$$

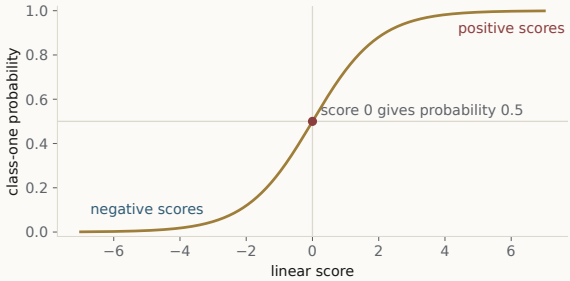
Here z is a real-valued score, not a probability.

Score z	-2	0	2
Probability $\sigma(z)$	0.119	0.500	0.881

Main idea

Higher scores imply higher class-one probabilities, always between zero and one.

Sigmoid Probability Curve



Negative scores
probability below 0.5

Zero score
probability 0.5

Positive scores
probability above 0.5

Takeaway. The sigmoid makes a linear score usable for probability prediction.

Logistic Probability Model

With one feature x, use a score

$$\underbrace{z(x)}_{\text{score}} = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} x.$$

Convert it into the class-one probability:

$$\boxed{p(x) = \frac{1}{1 + e^{-z(x)}}.}$$

β_0, β_1 are fitted coefficients. For multiple features, use $z(\boldsymbol{x}) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$, where p is the number of features.

Logistic Example: Exam Outcomes

Suppose an illustrative fitted model is

$$z(x) = -3 + 0.8x,$$

where x is hours of study and $Y = 1$ means passing.

Hours x	2	5	7
Score $z(x)$	-1.4	1.0	2.6
Probability $p(x)$	0.198	0.731	0.931

For 5 hours, for example,

$$\boxed{p(5) = \frac{1}{1 + e^{-1}} \approx 0.731.}$$

The output is a probability, not yet a pass/fail decision.

Logistic Decision Boundary

If incorrect predictions have equal costs, a common threshold is 0.5:

$$\widehat{y} = \begin{cases} 1, & p(x) \geq 0.5, \\ 0, & p(x) < 0.5. \end{cases}$$

In our example, $p(x) \geq 0.5$ precisely when $z(x) \geq 0$.
Thus

$$-3 + 0.8x \geq 0 \quad \Longleftrightarrow \quad \boxed{x \geq 3.75.}$$

At this threshold, the model predicts passing for $x \geq 3.75$ study hours.

Odds and Log-Odds

Motivation: logistic coefficients are easier to interpret through odds than through raw probability changes.
For event probability p, define

$$\boxed{\text{odds} = \frac{p}{1-p}}, \qquad \boxed{\text{log-odds} = \log \frac{p}{1-p}}.$$

If $p = 0.75$, then

$$\text{odds} = 3, \qquad \text{log-odds} = \log 3 \approx 1.099.$$

Odds of 3 describe a $3{:}1$ ratio of events to non-events, not a probability of 3.

Linear Model for Log-Odds

The sigmoid probability model is equivalent to

$$\boxed{\log \frac{p(x)}{1-p(x)} = \beta_0 + \beta_1 x.}$$

Thus logistic regression is **linear in log-odds**, not in probability.
The probability $p(x)$ changes nonlinearly with x, even when the right-hand side is linear.
This lets the model use familiar linear coefficients while predicting valid probabilities.

Logistic Coefficients as Odds Ratios

With all other features fixed, a one-unit increase in x adds β_1 to log-odds.
Therefore, the odds are multiplied by

$$\boxed{\text{odds ratio} = e^{\beta_1}.}$$

In our exam example, $\beta_1 = 0.8$:

$$\boxed{e^{0.8} \approx 2.23.}$$

An extra study hour multiplies modeled odds of passing by about 2.23.

Common mistake

This does *not* mean that the probability increases by 2.23 or by a fixed percentage.

Odds Ratios versus Probability Changes

For our fitted model $z(x) = -3 + 0.8x$, compare two one-hour increases.

Study hours x	4	5	7	8
Pass probability	0.550	0.731	0.931	0.968

From 4 to 5 hours, probability rises by approximately 0.181.
From 7 to 8 hours, it rises by only 0.037.

Main idea

Each extra hour multiplies the *odds* by $e^{0.8} \approx 2.23$, but the probability gain is not constant.

Motivation for Logistic Training

Earlier we were *given* the coefficients of an exam-passing model. Real training data contain labels, not the correct coefficients.
We need to fit coefficients that:

- assign high probability to cases labeled 1;
- assign low probability to cases labeled 0;
- strongly penalize confident mistakes.

Bernoulli likelihood measures how much probability a model assigns to the labels actually observed.

Bernoulli Likelihood for One Label

For training case i, let $y_i \in \{0, 1\}$ be its true label and $p_i = p(\boldsymbol{x}_i)$ its predicted class-one probability.
The probability assigned to the observed label is

$$\boxed{P(Y_i = y_i \mid \boldsymbol{x}_i) = p_i^{y_i}(1 - p_i)^{1-y_i}.}$$

Observed label	Likelihood contribution
$y_i = 1$	p_i
$y_i = 0$	$1 - p_i$

The exponents simply select the appropriate probability.

Likelihood: Correct and Incorrect Predictions

For two independent observations with true labels $\boldsymbol{y} = (1, 0)$, let p_i be the predicted probability of class 1 for case i.

Good predictions: $\boldsymbol{p} = (0.8, 0.2)$

$$L_{\text{good}} = 0.8\,(1 - 0.2) = \boxed{0.64}.$$

Confidently wrong predictions: $\boldsymbol{p} = (0.1, 0.9)$

$$L_{\text{wrong}} = 0.1\,(1 - 0.9) = \boxed{0.01}.$$

A higher likelihood means the model assigned more probability to the observed labels.

Binary Cross-Entropy Loss

With many independent training labels, the likelihood is a product of many small numbers. Taking the logarithm turns that product into a sum.
For $y_i \in \{0, 1\}$ and predicted class-one probability p_i, the **negative log-likelihood** is

$$\mathcal{L} = -\sum_{i=1}^{n} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

This is **binary cross-entropy**. We fit model coefficients by minimizing $\mathcal{L}$; a lower value means greater probability assigned to the observed labels.

Cross-Entropy: Numerical Example

Same two cases: true labels $(1, 0)$.
Good predictions: likelihood 0.64

$$-\log(0.64) = \boxed{0.446}.$$

Confidently wrong: likelihood 0.01

$$-\log(0.01) = \boxed{4.605}.$$

Main idea

Cross-entropy assigns a much larger penalty to confidently incorrect probabilities.

Motivation for Gradient Descent

Training goal: find coefficients that minimize logistic cross-entropy.
The sigmoid makes the loss depend nonlinearly on those coefficients, so we cannot use the usual ordinary least-squares coefficient formula.
Gradient descent updates the coefficients repeatedly:

$$\boxed{\text{new coefficients} = \text{old coefficients} - \rho\,(\text{gradient}).}$$

The gradient indicates how loss changes as coefficients change. The learning rate $\rho > 0$ controls the step size.

Logistic Gradient for One Observation

With one feature and score $z_i = \beta_0 + \beta_1 x_i$, the casewise cross-entropy has derivative

$$\frac{\partial \mathcal{L}_i}{\partial z_i} = p_i - y_i.$$

By the chain rule,

$$\boxed{\frac{\partial \mathcal{L}_i}{\partial \beta_0} = p_i - y_i, \qquad \frac{\partial \mathcal{L}_i}{\partial \beta_1} = (p_i - y_i)x_i.}$$

The main ingredient is **predicted probability minus actual label**, weighted by the feature.

Gradient Descent: Numerical Update

Take one training case $x = 2,\ y = 1$, starting at $\beta_0 = 0, \beta_1 = 0$.
The current prediction is $p = 0.5$, so

$$\left(\frac{\partial \mathcal{L}}{\partial \beta_0}, \frac{\partial \mathcal{L}}{\partial \beta_1}\right) = (-0.5, -1).$$

With learning rate $\rho = 0.1$, update to

$$\boxed{(\beta_0, \beta_1) = (0.05, 0.10).}$$

The new score is $0.05 + 0.10(2) = 0.25$, giving $\boxed{p \approx 0.562}$.
The probability of the observed positive label has increased.

Gradient for the Full Training Set

With n observations and p features, let $\boldsymbol{X}$ be the $n \times (p+1)$ matrix of training inputs, with a leading 1 in each row for the intercept.
Let $\boldsymbol{y}$ be the vector of true binary labels and $\boldsymbol{p}$ the vector of predicted class-one probabilities.
For the *summed* cross-entropy loss, the gradient is

$$\boxed{\nabla_{\boldsymbol{\beta}} \mathcal{L} = \boldsymbol{X}^{\top}(\boldsymbol{p} - \boldsymbol{y}).}$$

This is the sum of each observation's feature-weighted prediction error, used to update the coefficients.

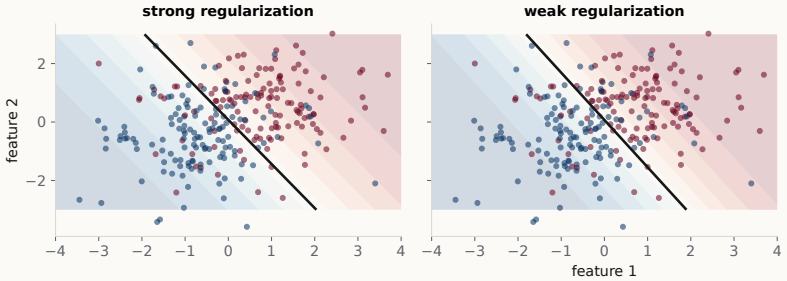
Motivation for Regularization

Motivation: with limited data, outliers, or highly correlated features, a logistic model may fit the training labels too closely.
Large coefficients may produce very extreme probabilities that do not generalize.
Regularization penalizes unnecessarily large coefficients during fitting.

Main idea

It can trade a little training fit for more stable predictions on new data.

Regularization and Decision Boundaries



Dots
training labels

Black boundary
probability 0.5

Background
predicted class probability

Takeaway. Regularization can stabilize a boundary influenced by unusual observations.

L2 Regularization

A common choice penalizes squared coefficients:

$$\mathcal{L}_\lambda = \mathcal{L} + \frac{\lambda}{2} \sum_{j=1}^{p} \beta_j^2.$$

- $\mathcal{L}$: binary cross-entropy from the training labels.
- $\lambda \geq 0$: penalty strength, selected using validation data.
- β_j: feature coefficients; the intercept β_0 is not penalized here.

Larger λ usually shrinks coefficients more strongly.

LDA and Logistic Regression Compared

LDA: learn Gaussian feature distributions *within* each class, with shared covariance. Apply Bayes' rule to obtain class probabilities.

Logistic regression: learn class-one probabilities *directly* by fitting linear log-odds with cross-entropy.

Both methods can produce a linear decision boundary, but they make different assumptions and learn parameters differently.

Main idea

Use validation performance and the plausibility of the assumptions to choose between them.

Evaluating classifiers

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Motivation for Decision Thresholds

Motivation: the same estimated probabilities may require different actions in screening, fraud detection, or routine classification.
Predict positive when $\widehat{p}(\boldsymbol{x}) \geq t$, with threshold t.
Consider six cases; the true labels and probabilities remain fixed:

Case	1	2	3	4	5	6
True label y_i	1	0	1	0	1	0
Predicted $\widehat{p}_i$	0.90	0.70	0.55	0.40	0.35	0.10

Compare the decisions made at $t = 0.50$ and $t = 0.30$.

Thresholds Change the Confusion Matrix

With $t = 0.50$, the predicted labels are

$$(1, 1, 1, 0, 0, 0).$$

With $t = 0.30$, they become

$$(1, 1, 1, 1, 1, 0).$$

Threshold	TP	FP	FN	TN
0.50	2	1	1	2
0.30	3	2	0	1

Lowering the threshold detects one more positive but creates one additional false alarm.

Thresholds Change Precision and Recall

From the previous six cases:

Threshold	Recall	Precision	Specificity
0.50	$2/3$	$2/3$	$2/3$
0.30	$3/3 = 1$	$3/5 = 0.60$	$1/3$

Lower t increases recall in this example, but precision and specificity fall.

Main idea

A threshold is a decision choice; it does not change the underlying estimated probabilities.

Class Imbalance and Accuracy

Suppose $1{,}000$ cases contain only 20 true positives.
One model gives

$$TP = 15, \quad FN = 5, \quad FP = 35, \quad TN = 945.$$

Then

$$\boxed{\text{Accuracy} = \frac{960}{1000} = 0.96,} \qquad \boxed{\text{Precision} = \frac{15}{50} = 0.30.}$$

A classifier predicting *negative for everyone* achieves 98% accuracy but detects **zero** positives.
High accuracy alone can be misleading with rare events.

ROC Curves and Error Trade-offs

A confusion matrix evaluates *one* threshold. To see how results change across thresholds, use a **ROC curve**.
It plots

$$\boxed{\mathrm{TPR} = \frac{TP}{TP + FN} \quad \text{(recall)}, \qquad \mathrm{FPR} = \frac{FP}{FP + TN}.}$$

For our six-case example:

Threshold	FPR	TPR
0.50	$1/3$	$2/3$
0.30	$2/3$	1

Each threshold produces one point on the ROC curve.

Precision–Recall Curves

When positives are uncommon, we need to know whether *positive predictions* are reliable as more positives are detected.
A **precision–recall (PR) curve** plots precision against recall as the threshold changes.

Threshold	Recall	Precision
0.50	$2/3$	$2/3$
0.30	1	$3/5 = 0.60$

Recall describes the fraction of actual positives found; precision describes the fraction of predicted positives that are correct.

ROC AUC Measures Ranking

The ROC curve contains many threshold choices. **ROC AUC** summarizes how well the scores rank positives above negatives overall.
For two positive scores $(0.9, 0.4)$ and two negative scores $(0.8, 0.2)$:

	vs. 0.8	**vs. 0.2**
Positive score 0.9	win	win
Positive score 0.4	loss	win

Three of four positive–negative comparisons are wins:

$$\boxed{\text{AUC} = 3/4 = 0.75.}$$

Ties count as half. AUC measures ranking, not calibration or the quality of a chosen threshold.

Motivation for Calibration

Motivation: a model can rank higher-risk cases correctly yet report probabilities that are too high or too low.
Calibration compares stated probabilities with observed frequencies.
If many similar cases receive predicted probability near 0.80, a well-calibrated model should see around 80% positives within that group.

Main idea

Ranking concerns *which cases are higher risk*. Calibration concerns *whether a stated probability is trustworthy*.

Calibration: Numerical Example

Consider two groups of 100 cases, all assigned probabilities near 0.80.

	Group A	**Group B**
Reported probability	0.80	0.80
Observed positive count	$80/100$	$40/100$
Interpretation	matches here	overconfident here

Calibration is assessed over many probability ranges, not from just one group. A confusion matrix at one threshold cannot measure it.

Decision Costs and Threshold Choice

Motivation: which threshold should we use when missing a positive is more costly than a false alarm?
Return to the six-case example:

Threshold	False positives	False negatives
0.50	1	1
0.30	2	0

With cost 1 for each false positive and 10 for each false negative:

$$\boxed{\text{Cost at } 0.50 = 11, \qquad \text{cost at } 0.30 = 2.}$$

Under these assumed costs, $t = 0.30$ performs better on these six examples; choose thresholds using appropriate validation data, not the test set.

Validation and Unseen Test Data

Motivation: comparing too many models or thresholds on the final test data can produce misleadingly optimistic results.
A practical workflow is:

1. Fit candidate models on training data.
2. Use validation to select complexity, regularization, and thresholds.
3. Evaluate once on untouched test data.

When records are related, such as repeated visits from one patient, split by patient rather than mixing that person's visits across sets.

The final test set evaluates a completed decision rule; it should not select one.

Choosing a classifier

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LDA, QDA, and Logistic Regression

These methods differ in what they assume about the data:

- **LDA:** Gaussian features with one shared covariance; stable with relatively less data per class.
- **QDA:** Gaussian features with class-specific covariance; more flexible but needs more data.
- **Logistic regression:** a linear model for log-odds, trained using cross-entropy; does not assume Gaussian features.

Compare them using appropriate validation metrics, then select the decision threshold for the application's costs.

Classification Study: Clinical Screening

Scenario: a clinic must decide who needs follow-up screening.

1. Define the positive class, features, and consequences of errors.
2. Compare LDA, QDA, and logistic regression with suitable validation.
3. Check recall and false alarms, together with ranking and calibration.
4. Select a threshold for the required decision cost.

Main idea

The best classifier is not automatically the most flexible one; it is the one that meets the task's goals on new data.

Summary

Classification	predict probabilities, then choose actions
LDA / QDA	use Gaussian class models and Bayes' rule
Logistic regression	use a sigmoid and learn coefficients via cross-entropy
Evaluation	check errors, thresholds, ranking, and calibration

Good probabilities + appropriate decisions + valid evaluation.

Week 8: from class probabilities to responsible decisions

Questions and discussion